ECG Noise and Lead-Failure Stress-Test Corpus (v1) — Generation Report and Full Record Roster

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Date: 2026-07-13 · **Repository:** https://github.com/depolarised/artefaux (tag `v1.0.0`) · **Sources (every value below traces to these):** `recipes/corpus.yaml`, `recipes/source_ids/*.csv`, `manifest.json` (all committed), regenerated in-memory from `artefaux.corpus` + `artefaux.recipes` (`master_seed = 20260713`). Generator: `scripts/reports/generate_co`

> **Scope.** This is a *stress-test set, not a clinically representative cohort.* It deliberately over-samples failure; do not use it to estimate real-world prevalence or deployment accuracy.

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1. Objectives

This is a deterministic generator and reproducible corpus **definition** for stress-testing ECG signal-quality gates (the internal `signalguard`) and noise detectors (`noiseguard`). It does **not** redistribute derived signals: it ships code, recipes, labels, manifest, and provenance so a user regenerates the corpus **bit-exactly** from their own open PhysioNet copies.

This report documents the generation algorithm and enumerates **all 67 stress records** with their corruption truth and expected behaviour. Records resolve against a local PhysioNet mirror; the concrete PTB-XL parent ids shown here are those in `recipes/source_ids/` (reproducible, seeded selection).

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2. Corpus composition

Group	n	Source	Corruption
Naturally poor	15	PTB-XL technical-validation quality flags	none (inherently noisy)

Group	n	Source	Corruption
Real-noise pairs	30	clean PTB-XL + NSTDB `em`/`ma`/`bw`	SNR ladder {-6, +0, +6, +12, +18} dB
Engineering extremes	22	clean PTB-XL parents (+ MACECGDB motion)	single-lead (12) + multi-lead (10)
Total	**67**		11 labelled data-integrity failures

Each record carries three label layers — **clinical parent** (authored rhythm class + PTB-XL quality flags; the Uni-G statement field is reserved and empty in v1), **corruption truth** (electrodes/leads, requested + measured SNR, noise segment, seed, α, amplitude bookkeeping), and **expected behaviour** (per-lead and record-level, mapped to the `signalguard` `QualityLabel` / `RecordQuality` and `noiseguard` discard vocabularies).

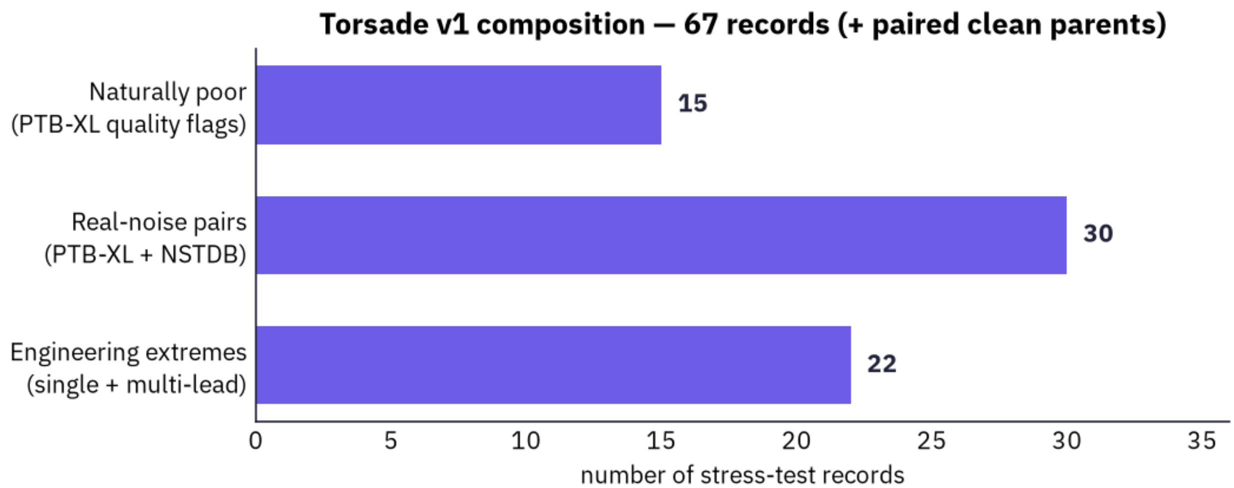

Figure 1: Corpus composition — 67 stress records plus paired clean parents.

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3. Sources and licensing

Dataset	Role	Version	License
PTB-XL	clean + naturally-poor parents (500 Hz)	1.0.3	CC-BY-4.0
PTB-XL+	Uni-G/Glasgow layer — *cited and pinned, not consumed by v1*	1.0.1	CC-BY-4.0
MIT-BIH NSTDB	`em`/`ma`/`bw` noise for the SNR ladder	1.0.0	ODC-BY-1.0
MACECGDB	real standing/walking/jumping motion	1.0.0	ODC-BY-1.0

PTB-XL, PTB-XL+, and MACECGDB are read from a local PhysioNet mirror; **only NSTDB** is fetched by `make download`. **Code** is GPL-3.0-or-later; **manifest, labels, and docs** are CC-BY-4.0. No source signals are redistributed — see `ATTRIBUTION.md`.

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4. Generation algorithm

4.1 Canonical form. Every parent becomes a `(12, T)` float64 mV array in canonical lead order `[I, II, III, aVR, aVL, aVF, V1-V6]` at 500 Hz (NSTDB resampled from 360 Hz by deterministic polyphase resampling). Absent limb leads are derived from I and II.

4.2 Real-noise SNR mixing. For a clean lead x, real noise n, and requested SNR s (dB): $\alpha = \sqrt{P_x/(P_n \cdot 10^{s/10})}$, $y = x + \alpha\, n$, with P_x, P_n the DC-removed mean-square powers over the contaminated interval only. Both the requested and the *measured* SNR, and the exact NSTDB segment offset, are recorded in the label.

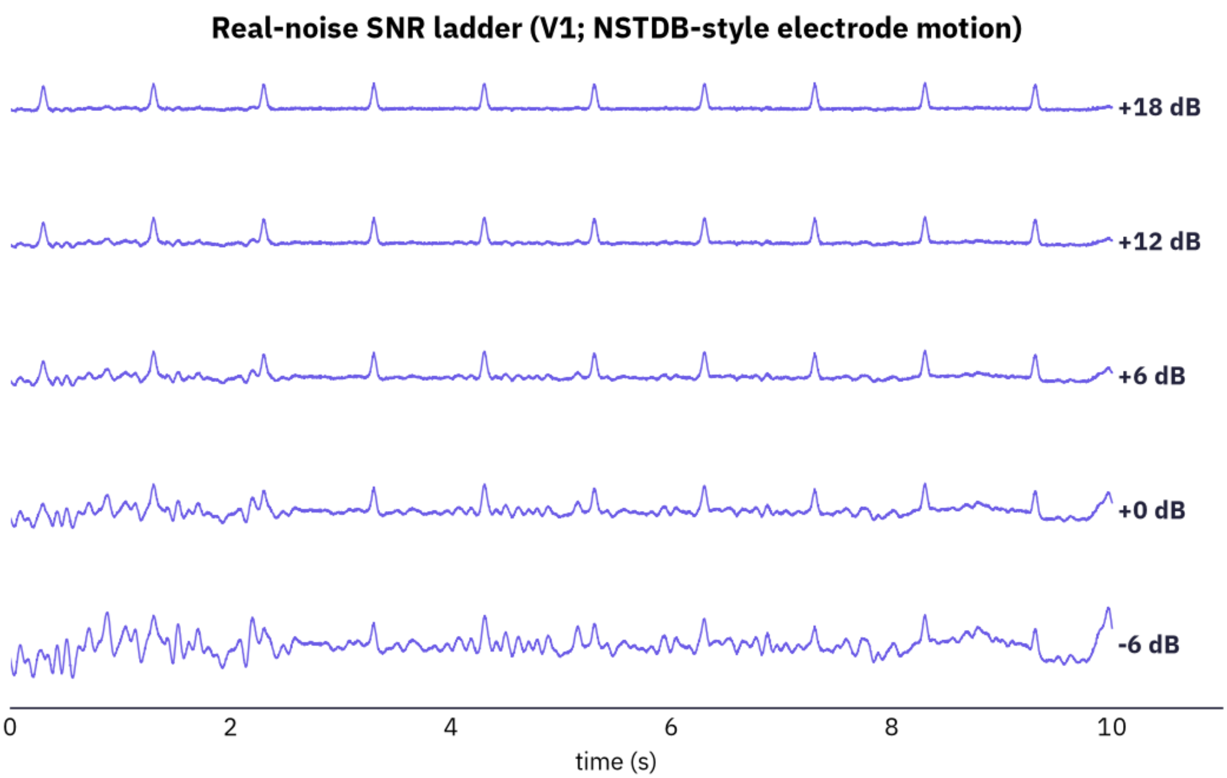

Figure 2: Real-noise SNR ladder (illustrative synthetic parent; NSTDB-style electrode motion).

4.3 Electrode-domain corruption. Limb-electrode potentials are recovered in the RA-as-reference gauge (`RA = 0`, `LA = I`, `LL = II`); chest-electrode potentials as `Ck = Vk + WCT`, `WCT = (RA+LA+LL)/3`. An artefact is added to a chosen electrode; the WCT and all twelve leads are recomputed, and the invalidated derived leads are recorded. (The classical WCT is an idealised electrode model, **not** a torso forward model.)

4.4 Engineering extremes. Deterministic builders for clipping/rail-saturation, flatline (zero), constant-ADC, lead-off, digital-missing (NaN), electrode-motion swings, step-and-exponentialrecovery, polarity inversion, and intermittent lead-off. NaN / flat / constant / rail cases are labelled `data_integrity_failure` with a type — **not** as noise.

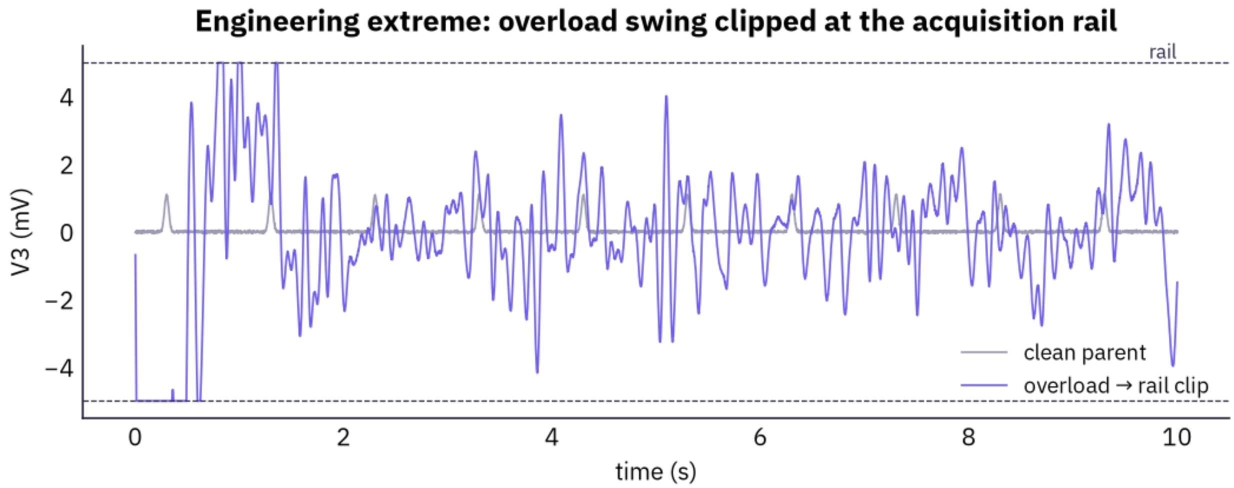

Figure 3: Engineering extreme: an overload swing clipped at the acquisition rail.

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5. Full record roster

All 67 records are listed below. **Expected quality** is the record-level `RecordQuality` a correctly-tuned gate should assign; **Discard** is the expected `noiseguard` record-level decision; **Bad leads** is the count of leads the label expects a detector to reject. PTB-XL parents are shown as record stems (the `records500/NNNNN/` directory prefix is omitted).

5.1 Naturally poor (15)

Real PTB-XL records selected by technical-validation quality flags; no clean parent (inherently bad).

Record	PTB-XL source	Quality flag(s)	Expected quality	Discard
`nlf_nat_001`	`10646_hr`	electrodes_problems	**reject**	yes
`nlf_nat_002`	`10652_hr`	burst_noise	**limited**	—
`nlf_nat_003`	`11176_hr`	static_noise	**limited**	—
`nlf_nat_004`	`12587_hr`	static_noise	**limited**	—
`nlf_nat_005`	`13777_hr`	static_noise	**limited**	—
`nlf_nat_006`	`14197_hr`	static_noise	**limited**	—
`nlf_nat_007`	`16149_hr`	baseline_drift	**reject**	yes
`nlf_nat_008`	`16981_hr`	static_noise	**limited**	—
`nlf_nat_009`	`17281_hr`	static_noise	**limited**	—
`nlf_nat_010`	`17293_hr`	baseline_drift	**limited**	—
`nlf_nat_011`	`19487_hr`	static_noise	**limited**	—
`nlf_nat_012`	`20478_hr`	static_noise	**limited**	—
`nlf_nat_013`	`00359_hr`	static_noise	**reject**	yes
`nlf_nat_014`	`06320_hr`	baseline_drift	**limited**	—
`nlf_nat_015`	`06968_hr`	static_noise	**limited**	—

5.2 Real-noise SNR-ladder pairs (30)

Clean PTB-XL parents mixed with NSTDB noise across the ladder {-6, +0, +6, +12, +18} dB. Each ships with its untouched clean parent (`*_clean`).

Record	Rhythm	PTB-XL parent	Noise @ SNR	Leads	Bad leads	Expected quality	Discard
`nlf_noise_001`	sinus	`10236_hr`	`em` @ -6 dB	12	12	**reject**	yes
`nlf_noise_002`	afib	`10330_hr`	`ma` @ +0 dB	6	0	**limited**	yes
`nlf_noise_003`	pac pvc	`11089_hr`	`bw` @ +6 dB	6	0	**limited**	—
`nlf_noise_004`	bbb conduction	`11662_hr`	`em` @ +12 dB	12	0	**diagnostic**	—
`nlf_noise_005`	st t	`12086_hr`	`ma` @ +18 dB	6	0	**diagnostic**	—
`nlf_noise_006`	flutter svt	`12407_hr`	`bw` @ -6 dB	6	6	**limited**	yes
`nlf_noise_007`	sinus	`01306_hr`	`em` @ +0 dB	12	0	**limited**	yes
`nlf_noise_008`	afib	`13114_hr`	`ma` @ +6 dB	6	0	**limited**	—
`nlf_noise_009`	pac pvc	`13332_hr`	`bw` @ +12 dB	6	0	**diagnostic**	—
`nlf_noise_010`	bbb conduction	`13611_hr`	`em` @ +18 dB	12	0	**diagnostic**	—
`nlf_noise_011`	st t	`14071_hr`	`ma` @ -6 dB	6	6	**limited**	yes
`nlf_noise_012`	flutter svt	`14211_hr`	`bw` @ +0 dB	6	0	**limited**	yes
`nlf_noise_013`	sinus	`15473_hr`	`em` @ +6 dB	12	0	**limited**	—
`nlf_noise_014`	afib	`15535_hr`	`ma` @ +12 dB	6	0	**diagnostic**	—
`nlf_noise_015`	pac pvc	`15877_hr`	`bw` @ +18 dB	6	0	**diagnostic**	—

Record	Rhythm	PTB-XL parent	Noise @ SNR	Leads	Bad leads	Expected quality	Discard
`nlf_noise_016`	bbb conduction	`16615_hr`	`em` @ -6 dB	12	12	**reject**	yes
`nlf_noise_017`	st t	`16930_hr`	`ma` @ +0 dB	6	0	**limited**	yes
`nlf_noise_018`	flutter svt	`17553_hr`	`bw` @ +6 dB	6	0	**limited**	—
`nlf_noise_019`	sinus	`17727_hr`	`em` @ +12 dB	12	0	**diagnostic**	—
`nlf_noise_020`	afib	`17770_hr`	`ma` @ +18 dB	6	0	**diagnostic**	—
`nlf_noise_021`	pac pvc	`17867_hr`	`bw` @ -6 dB	6	6	**limited**	yes
`nlf_noise_022`	bbb conduction	`18284_hr`	`em` @ +0 dB	12	0	**limited**	yes
`nlf_noise_023`	st t	`18508_hr`	`ma` @ +6 dB	6	0	**limited**	—
`nlf_noise_024`	flutter svt	`19002_hr`	`bw` @ +12 dB	6	0	**diagnostic**	—
`nlf_noise_025`	sinus	`19238_hr`	`em` @ +18 dB	12	0	**diagnostic**	—
`nlf_noise_026`	afib	`01979_hr`	`ma` @ -6 dB	6	6	**limited**	yes
`nlf_noise_027`	pac pvc	`20943_hr`	`bw` @ +0 dB	6	0	**limited**	yes
`nlf_noise_028`	bbb conduction	`20977_hr`	`em` @ +6 dB	12	0	**limited**	—
`nlf_noise_029`	st t	`21495_hr`	`ma` @ +12 dB	6	0	**diagnostic**	—
`nlf_noise_030`	flutter svt	`21820_hr`	`bw` @ +18 dB	6	0	**diagnostic**	—

5.3 Engineering extremes — single-lead (12)

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
`nlf_eng_001_emg_v1`	`02965_hr`	swing 6 mV p2p (V1)	—	1	**limited**	—
`nlf_eng_002_macecg_walk_v2`	`03016_hr`	MACECGDB `walk` swing 8 mV p2p (V2)	—	1	**limited**	—
`nlf_eng_003_intermittent_v6`	`03047_hr`	intermittent lead-off with reconnection (V6)	intermittent lead off	1	**limited**	—
`nlf_eng_004_step_ii`	`03365_hr`	baseline step 4 mV, tau=1 s (II)	—	0	**limited**	—
`nlf_eng_005_macecg_jump_overload_v3`	`03427_hr`	MACECGDB `jump` swing 30 mV p2p (V3) $\rightarrow$ rail $\pm$5 mV	rail saturation	1	**limited**	—
`nlf_eng_006_flatline_v4`	`03543_hr`	flatline / zero-fill (V4)	flatline zero	1	**limited**	—
`nlf_eng_007_constant_v5`	`03687_hr`	stuck constant 1.2 mV (V5)	constant adc	1	**limited**	—
`nlf_eng_008_missing_v6`	`03730_hr`	digital-missing / NaN (V6)	digital missing channel	1	**limited**	—
`nlf_eng_009_flatline_i`	`03941_hr`	flatline / zero-fill (I)	flatline zero	5	**limited**	—
`nlf_eng_010_leadoff_v2`	`04593_hr`	lead-off (V2)	lead off	1	**limited**	—
`nlf_eng_011_emg_burst_v5`	`00476_hr`	swing 5 mV p2p (V5), t in [3,6] s	—	0	**limited**	—
`nlf_eng_021_reversal_i`	`09509_hr`	polarity inversion (I)	—	0	**limited**	—

5.4 Engineering extremes — multi-lead / mixed (10)

Electrode-domain propagation, precordial coupling, multi-lead saturation, and compound "wild" cases.

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
`nlf_eng_012_couple_v1v2`	`05267_hr`	precordial coupling V1, V2 @ +0 dB	—	2	**limited**	—
`nlf_eng_013_couple_v4v5`	`05298_hr`	precordial coupling V4, V5 @ +6 dB	—	0	**limited**	—
`nlf_eng_014_triple_v1v2v3`	`05887_hr`	precordial coupling V1, V2, V3 @ -6 dB	—	3	**limited**	yes
`nlf_eng_015_electrode_la`	`06175_hr`	LA electrode motion 1 mV	—	4	**limited**	yes
`nlf_eng_016_electrode_ra`	`06526_hr`	RA electrode offset 3 mV	—	6	**reject**	yes
`nlf_eng_017_saturate_v2v3`	`06789_hr`	overload swing 30 mV p2p (V2, V3) → rail ±5 mV	rail saturation	2	**limited**	—
`nlf_eng_018_saturate_v4v5v6`	`07086_hr`	overload swing 30 mV p2p (V4, V5, V6) → rail ±5 mV	rail saturation	3	**limited**	yes
`nlf_eng_019_electrode_ll`	`08699_hr`	LL electrode motion 1.2 mV	—	3	**limited**	yes
`nlf_eng_020_compound_wild`	`08931_hr`	LA electrode motion 0.8 mV ; MACECGDB `walk` swing 8 mV p2p (V2) ; lead-off (V5)	lead off	5	**limited**	yes

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
`nlf_eng_022_mixed_integrity`	`09915_hr`	digital-missing / NaN (V1) ; overload swing 25 mV p2p (V6) → rail ±5 mV ; baseline step 3 mV, tau=1.5 s (II)	digital missing channel	2	**limited**	—

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6. Reproducibility and provenance

A single master seed (`20260713`) plus `provenance.json` (pinned source versions, per-record seed, source-file SHA-256) make the corpus bit-exactly regenerable:

```
make download        # NSTDB only (PTB-XL / PTB-XL+ / MACECGDB from the local mirror)
make select          # resolve PTB-XL parent ids -> recipes/source_ids/  (seeded)
make regenerate      # WFDB records + labels + manifest + provenance -> ./out
```

Determinism is per-record: `SeedSequence(entropy=master_seed, spawn_key=(record_index,))`. The corpus definition leaves source ids unresolved; `resolve_source_ids()` binds the seeded PTB-XL selection into the specs at generation time.

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7. Verification

Confirmed end-to-end on the local PhysioNet mirror (PTB-XL 1.0.3, MACECGDB 1.0.0, NSTDB 1.0.0):

- **Counts:** 67 records (15 / 30 / 22) + 52 paired clean parents.
- **SNR ladder:** max |requested - measured| SNR error **0.0 dB** across all 30 mixes (tol 0.5 dB).
- **Data integrity:** digital-missing (NaN) round-trips through WFDB — the affected channel reads back all-NaN while intact leads stay finite.
- **Determinism:** two independent full regenerations are **byte-identical**.
- **Quality gates:** 61 unit/integration tests pass; `ruff` + `black` clean.

Generated by Artefaux v1.0.0 (https://github.com/depolarised/artefaux) via `scripts/reports/generate_corpus_` *every figure and table value is computed from the committed corpus definition.*